

Cooling Equipment HMI Application Interface Specification

Pentamaster 5-13-2024



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1.0 Introduction

The Human Machine Interface (HMI) system designed for managing tray or tube carriers post-baking process. This manual serves as a comprehensive guide to understanding and utilizing the features and functionalities of the HMI system effectively. Below is an overview of the key aspects covered in this manual:

1. Purpose of the HMI System

The HMI system is specifically tailored to streamline the management of tray or tube carriers after the baking process. It provides users with a centralized control interface for organizing and monitoring inventory within the storage area.

2. Integration of ATOP Pick To Light Technology

To enhance operational efficiency, the HMI system incorporates ATOP Pick To Light technology. This integration enables users to easily locate and retrieve carriers from the storage area, optimizing inventory management processes.

3. Implementation with Secs Gem Communication Protocol

Implemented with Secs Gem (SEMI Equipment Communications Standard/Generic Equipment Model), the HMI system ensures seamless communication and interoperability with other equipment and systems in the manufacturing environment. This standardized communication protocol facilitates efficient data exchange and integration.

4. Multi-page Interface Structure

The HMI system features a user-friendly interface comprising various pages tailored to different operational requirements. From the Main Page providing system overview to specialized pages such as Operator, Lot Status, Loading, Unloading, Secs Gem, Setting, and Pick-To-Light, users have access to comprehensive controls and functionalities.

1.1 Prerequisites

Before utilizing the HMI system, users should ensure the following prerequisites are met:

Secs Gem Dongle Key: A Secs Gem dongle key is required for secure communication between the HMI system and other equipment. Ensure the dongle key is properly inserted into the designated port before initiating system operation.

Windows PC: The HMI system is compatible with Windows-based PCs. Ensure that your PC meets the minimum system requirements specified by the manufacturer for optimal performance.



2.0 User Interface Overview

The HMI system provides users with a user-friendly interface to navigate and access various functionalities efficiently. Below are the key components of the interface:

A. Menu Button:

Function: Displays all available pages within the system, allowing users to navigate to different sections easily.

B. Alarm Button:

Function: Displays all active alarms and warnings, providing users with immediate notification of any system issues or abnormalities.

C. Sign In Button:

Function: Allows users to sign in to the system, providing access to personalized settings and functionalities based on user roles and permissions.

D. Lot Status Information:

Function: Provides real-time information on the number of running lots and open lots, enabling users to monitor production status at a glance.

E. Tray Bundles Rack Information:

Function: Displays detailed information about the tray bundles rack, including the percentage of location used, loaded shelves, and empty shelves.

F. Tubes Carrier Rack Information:

Function: Displays detailed information about the tubes carrier rack, including the percentage of location used, loaded shelves, and empty shelves.

G. Rack Indicator:

Function: Indicates the status of the racks within the machine layout, providing visual cues to users about the availability of storage space.

H. Reset Button:

Function: Clears all active alarms and warnings, allowing users to reset the system and resume normal operation.

I. Racking Image:

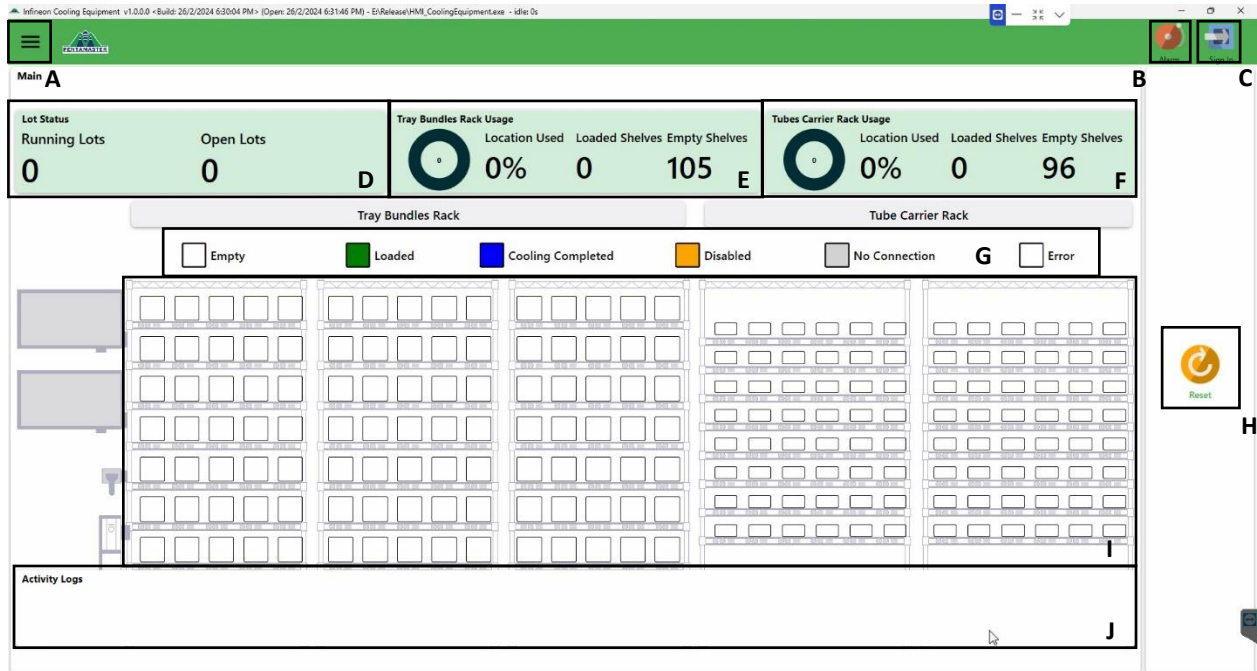
Function: Displays an image of the rack layout for tubes and trays, providing users with a visual representation of the storage configuration.



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J. Activity Logs:

Function: Displays a comprehensive log of all activities performed within the HMI system, including user interactions, system events, and operational changes. This log provides users with purposes.





3.0 Features

The HMI system offers various features to enhance user experience and streamline operations. Below are some of the key features:

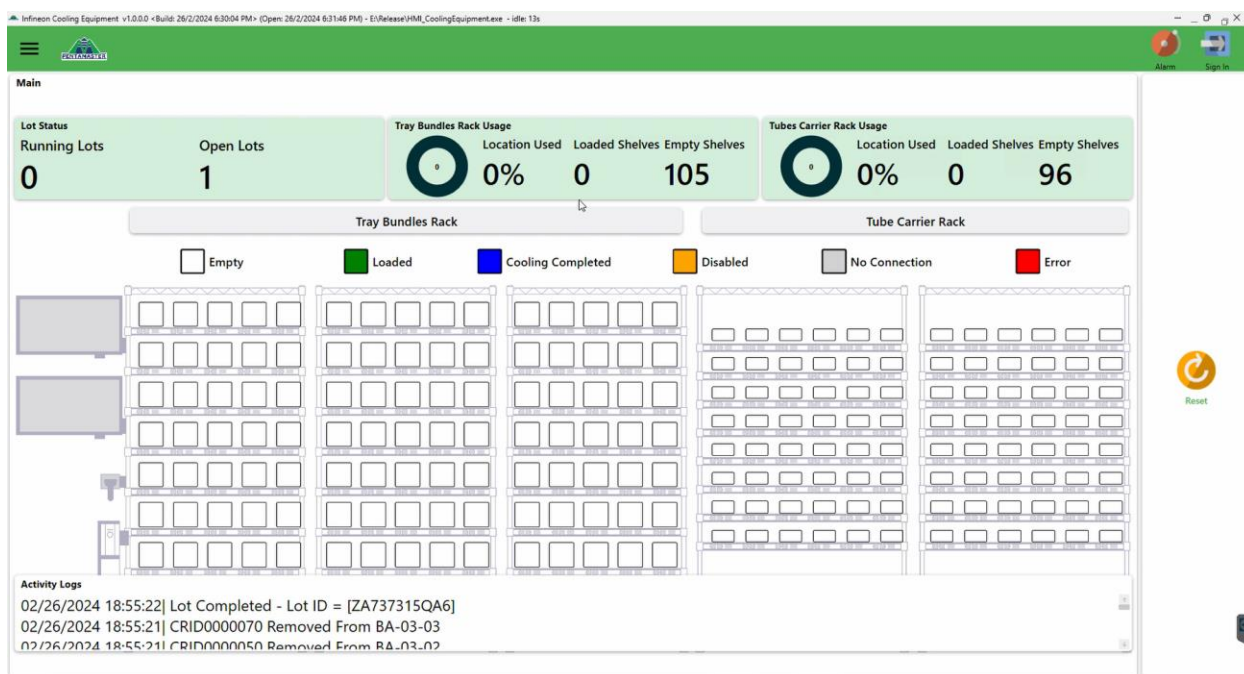
3.1 User Account Management

This section provides comprehensive guidance on managing user accounts and access permissions within the HMI interface. From signing in and out of the system to creating and editing user account groups, this documentation will walk you through the essential steps to efficiently manage user accounts. Whether you're an operator or an administrator, this guide will help you navigate the user account management features effectively, ensuring smooth operation of the HMI system.

3.1.1 User Account Sign In

To access the HMI system, users can follow these steps:

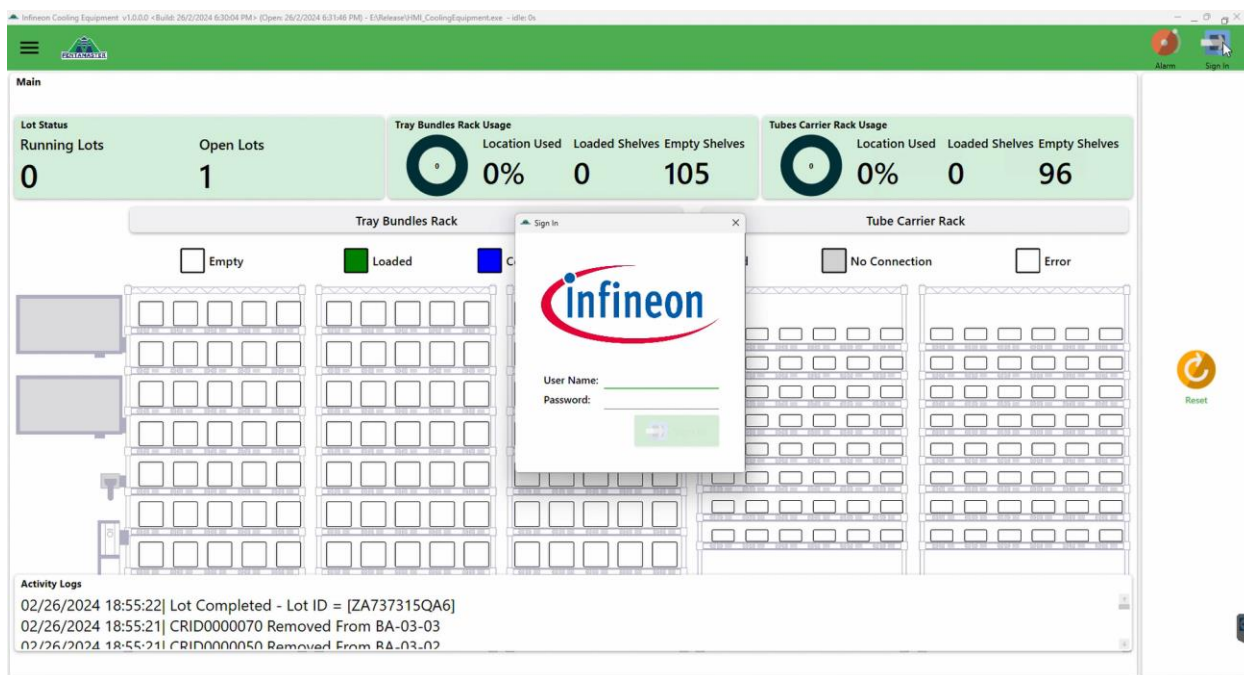
1. Click on the "Sign In" button located at the top right of the interface.



2. Enter the username and password in the provided fields within the pop-up window.



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3. Click on the "Sign In" button within the pop-up window to complete the sign-in process.

Notes:

- a. If the username is entered using a barcode scanner, the system will automatically log in without requiring the user to input their password.

Upon successful sign-in, users will be redirected to the main page. Depending on the user level, the main page may display different features and functionalities. Refer to the image below for an example of the main page for the "Operator" user level.

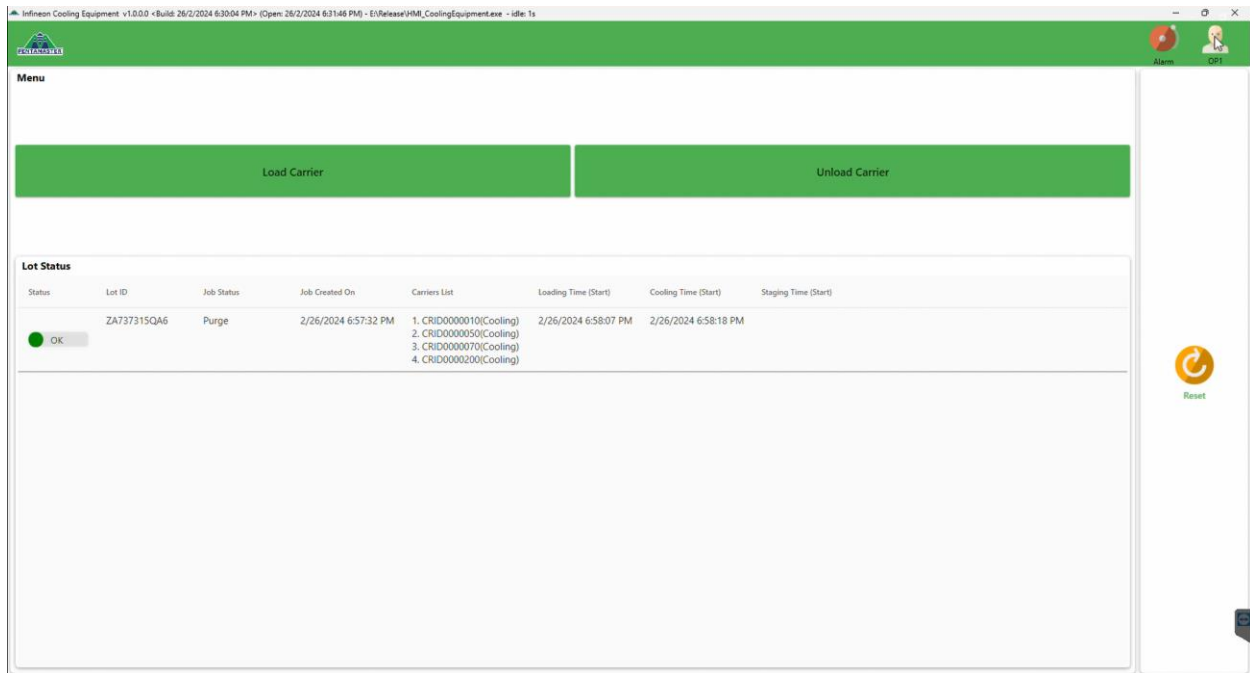
3.1.2 User Sign Out

To log out of the HMI system, users can follow these steps:

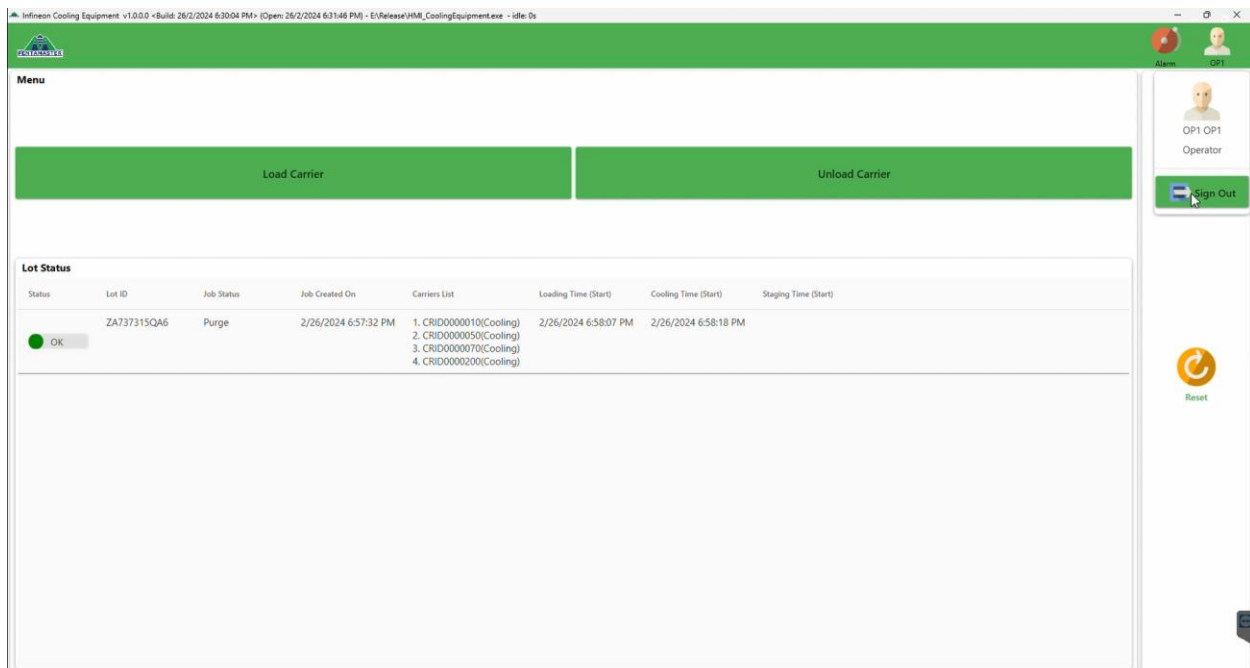
1. Click on the "Profile" button located at the top right of the interface.



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2. Within the pop-up window, click on the "Sign Out" button located at the bottom.



3. The system will automatically redirect users to the main page upon signing out.

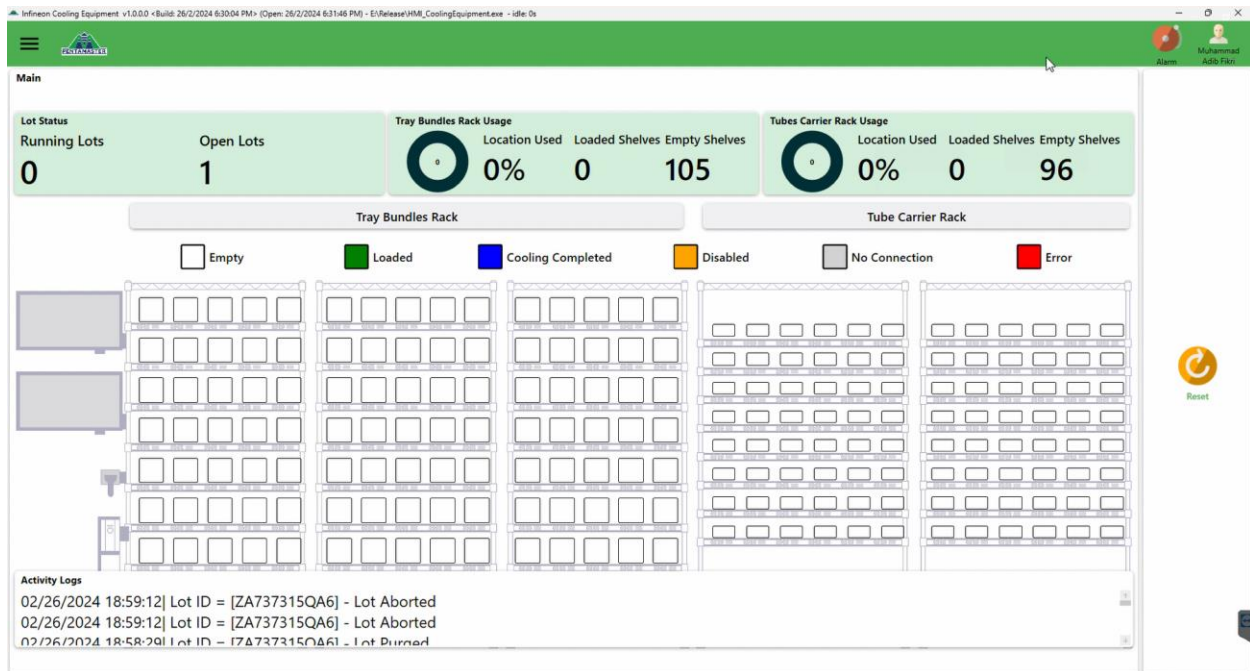
3.1.3 Add User Account Group

To add a new user account group in the HMI system, follow these steps:

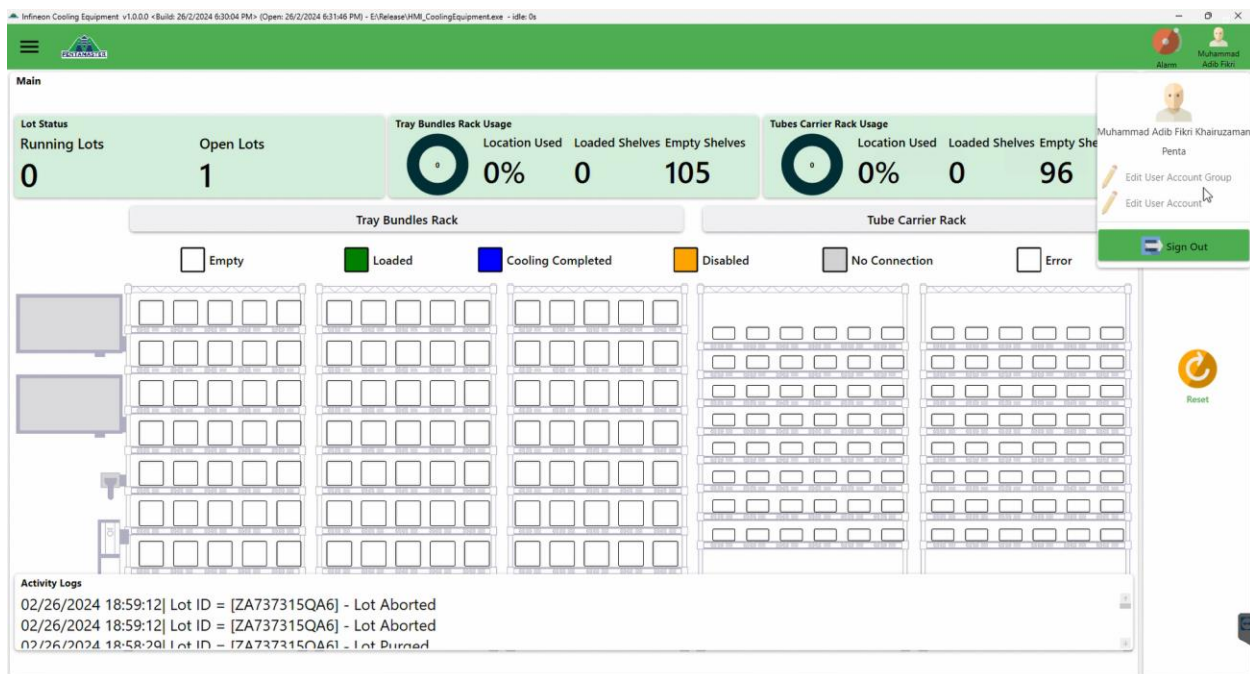


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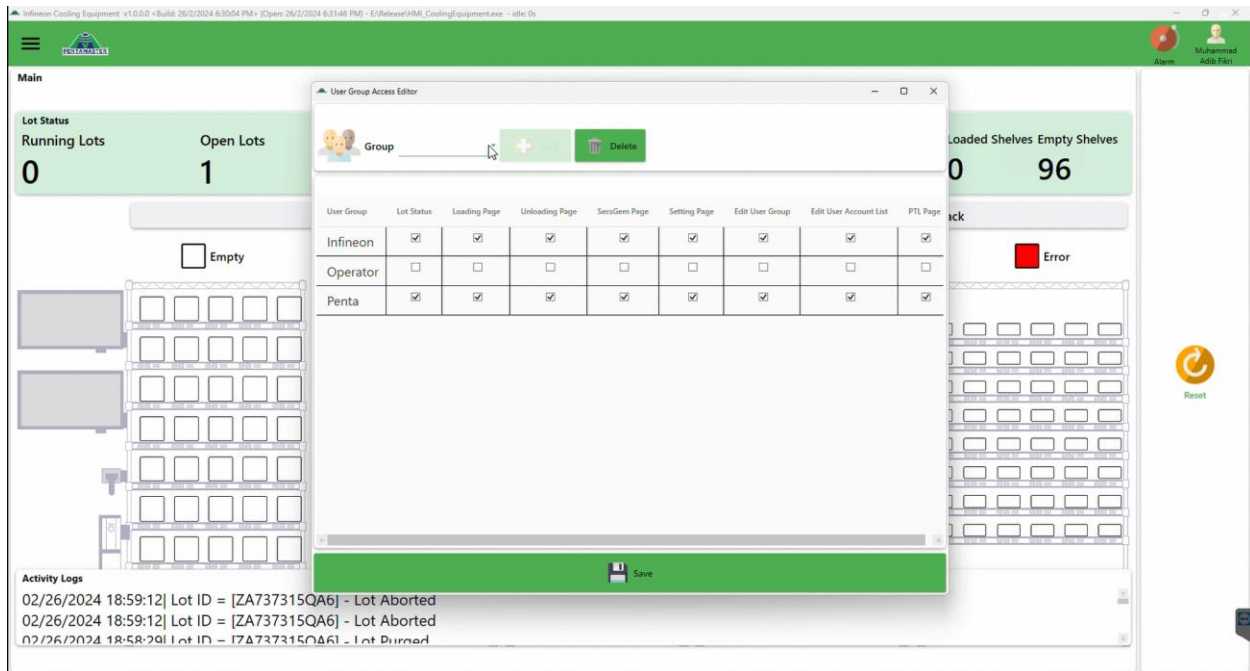
1. Sign in to the HMI system.
2. Click on the "Profile" button located at the top right of the interface.



3. From the pop-up window, select "Edit User Account Group."



4. In the provided blank space, type the name of the new group.



5. Click on the "Add" button to confirm the addition.
6. Users can modify the access permissions for the newly created group, specifying which pages can be accessed.
7. Once all modifications are complete, click on the "Save" button to apply the changes.

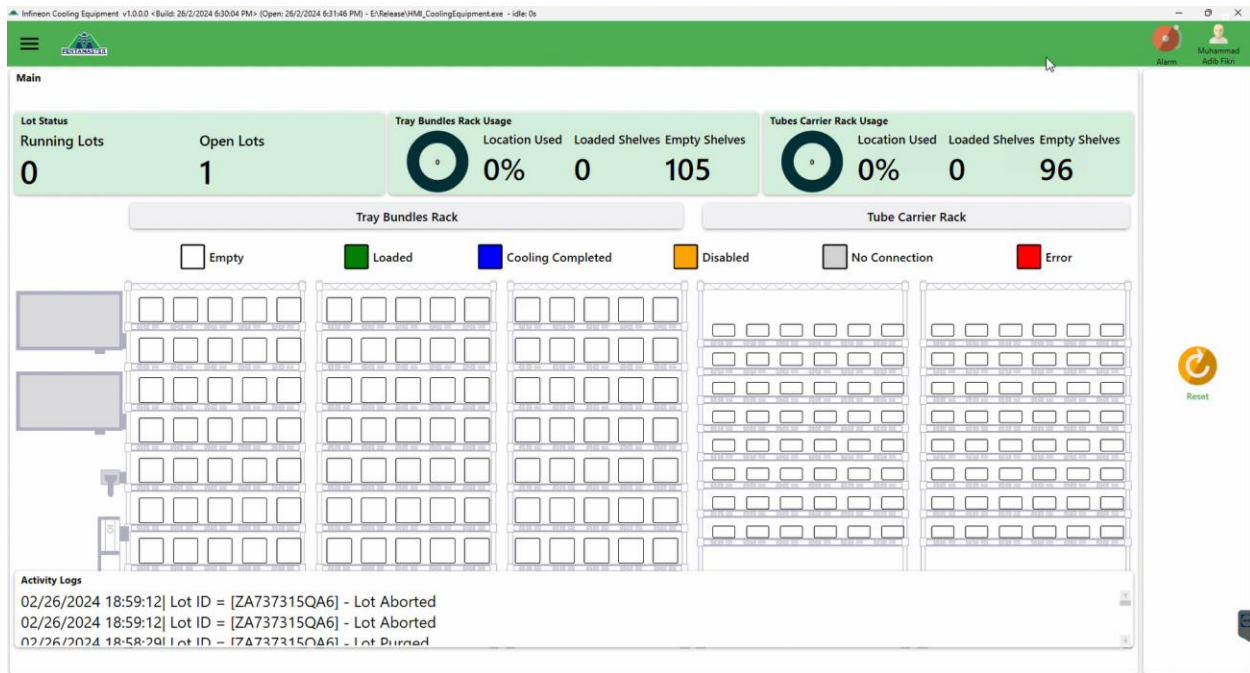
3.1.4 Edit User Account Group

To edit an existing user account group in the HMI system, follow these steps:

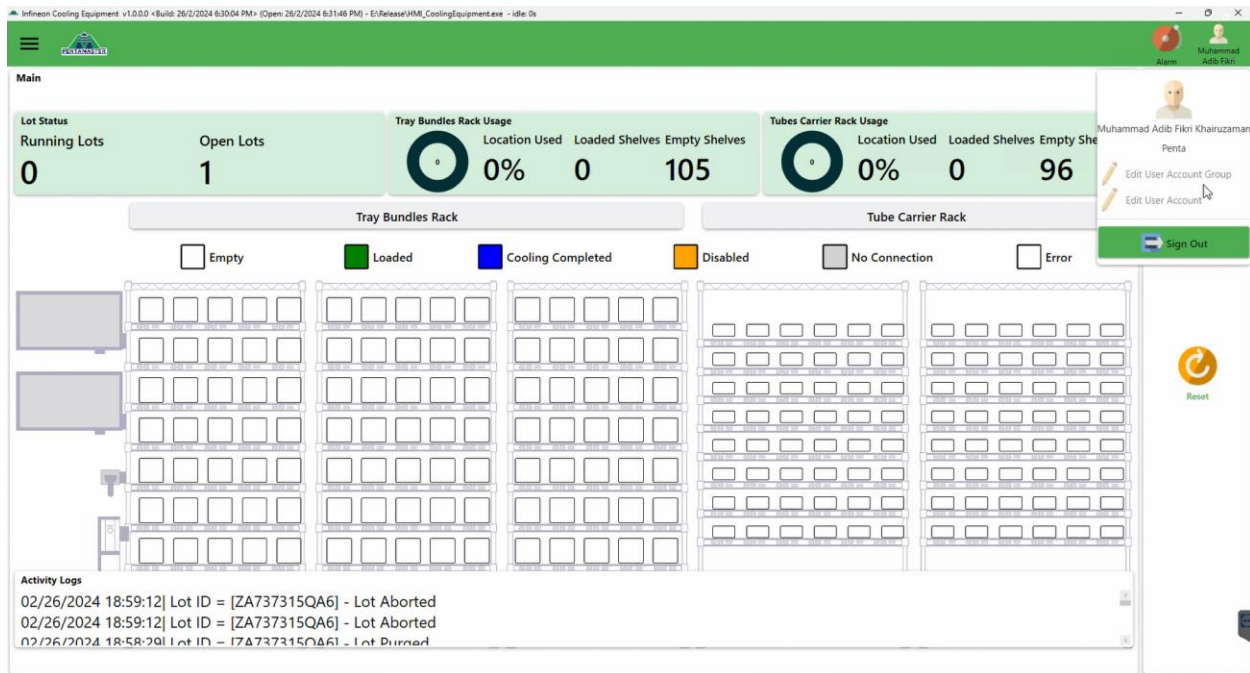
1. Sign in to the HMI system.
2. Click on the "Profile" button located at the top right of the interface.



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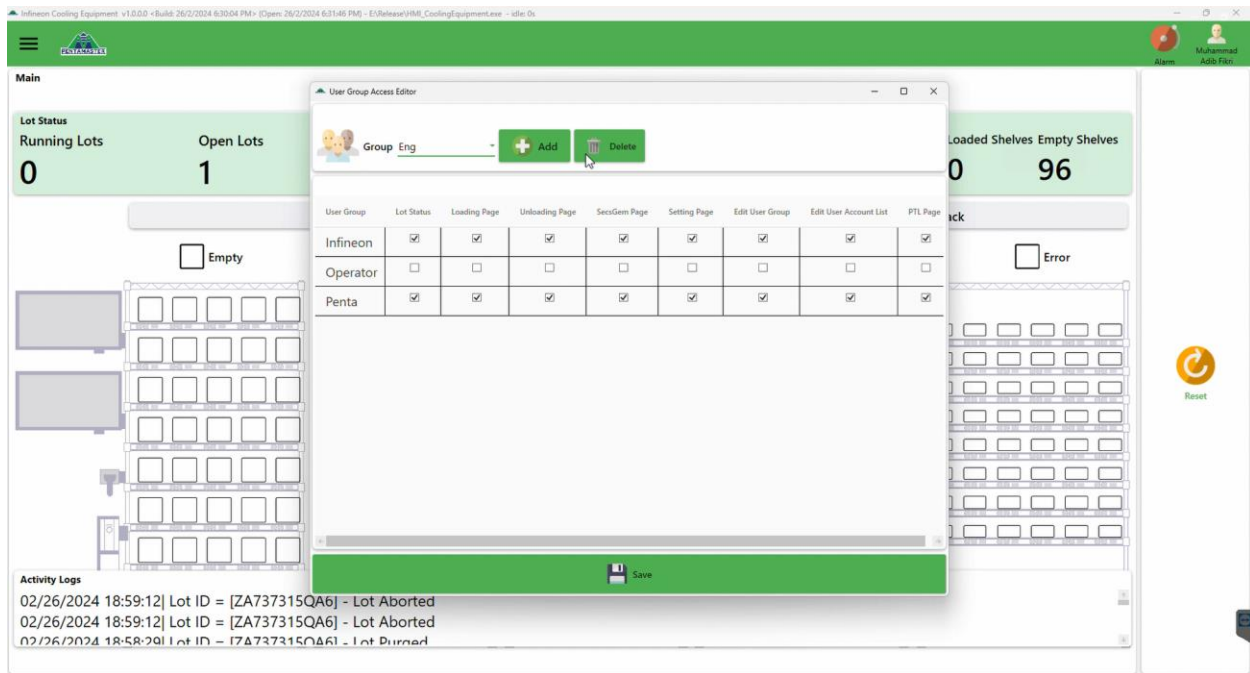


3. From the pop-up window, select "Edit User Account Group."

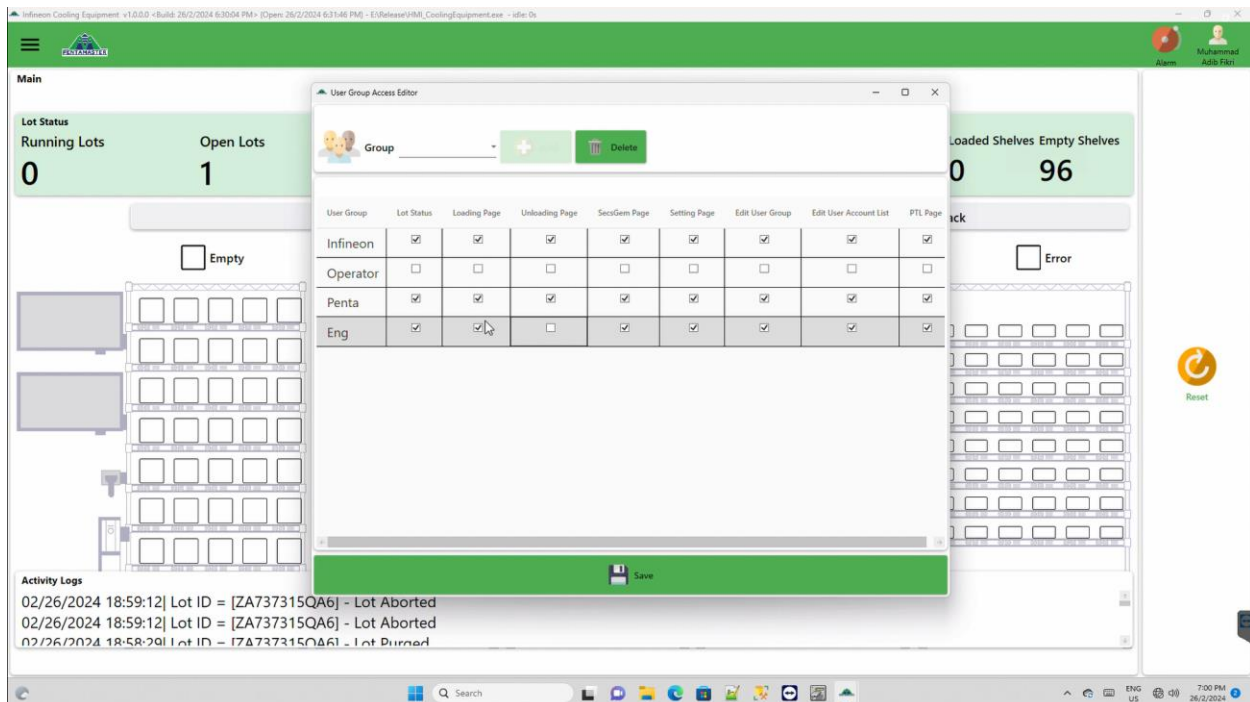


4. To delete a group, choose the desired group from the drop-down menu.

5. Click on the "Delete" button to remove the selected group.



6. Users can modify the access permissions for the selected group, specifying which pages can be accessed.



7. Once all modifications are complete, click on the "Save" button to apply the changes.

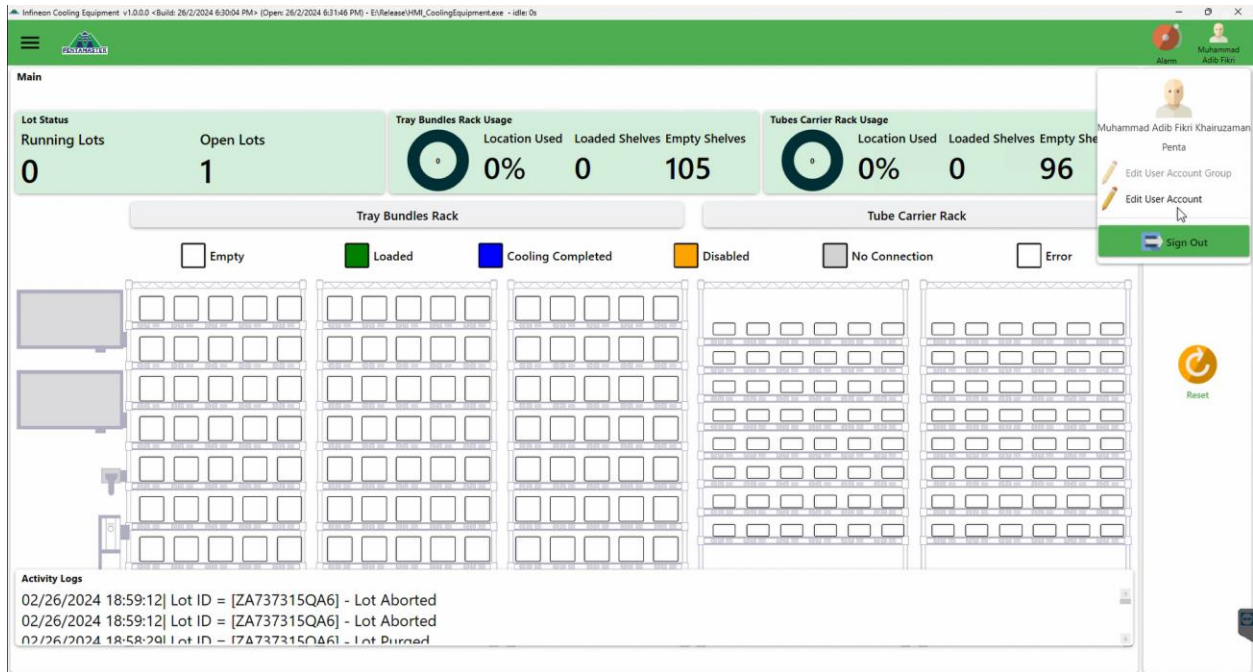
3.1.5 Adding a User Account



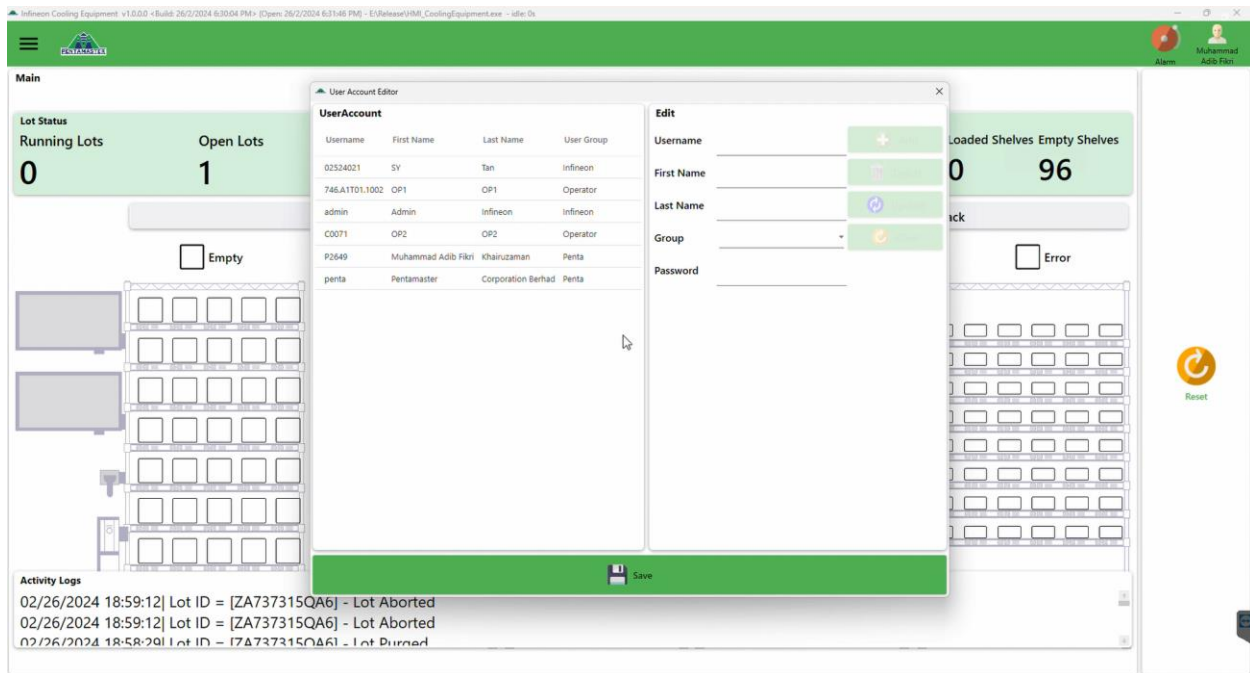
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To create a new user account in the HMI system, follow these steps:

1. Sign in to the HMI system.
2. Click on the "Profile" button located at the top right of the interface.
3. From the pop-up window, select "Edit User Account."



4. In the provided blank space, enter the username, first name, last name, group, and password for the new user.

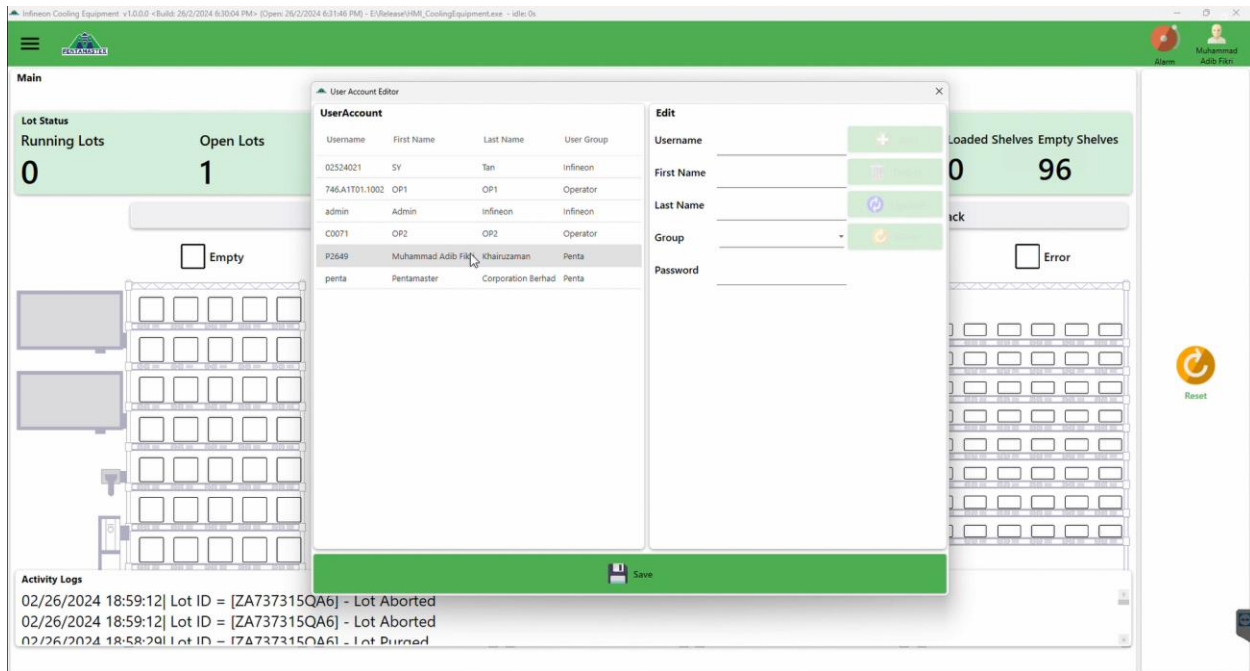


5. Click on the "Add" button to confirm the addition.
6. Once all modifications are complete, click on the "Save" button to apply the changes.

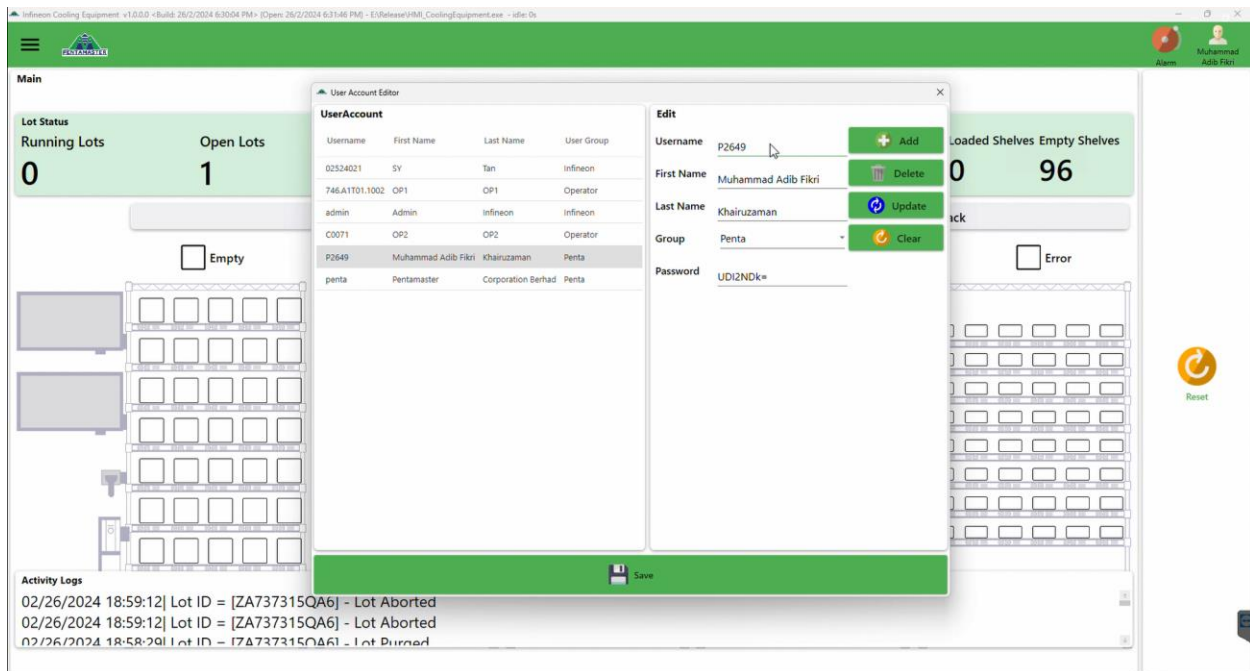
3.1.6 Editing a User Account

To modify an existing user account in the HMI system, follow these steps:

1. Sign in to the HMI system.
2. Click on the "Profile" button located at the top right of the interface.
3. From the pop-up window, select "Edit User Account."
4. To delete a user account, choose the desired account from the account list.



5. Click on the "Delete" button to remove the selected user.



6. Users can edit the user information for the selected user, such as username, first name, last name, and group.

7. Once all modifications are complete, click on the "Save" button to apply the changes.

3.2 Operational



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This section offers comprehensive guidance on the loading and unloading procedures within the HMI interface. From initiating loading sequences to overseeing the unloading process, this documentation will navigate users through the critical steps required to operate the HMI system efficiently during carrier handling operations. Whether you're fulfilling the role of a system operator or administrator, this guide will aid in effectively managing carrier activities within the cooling system environment.

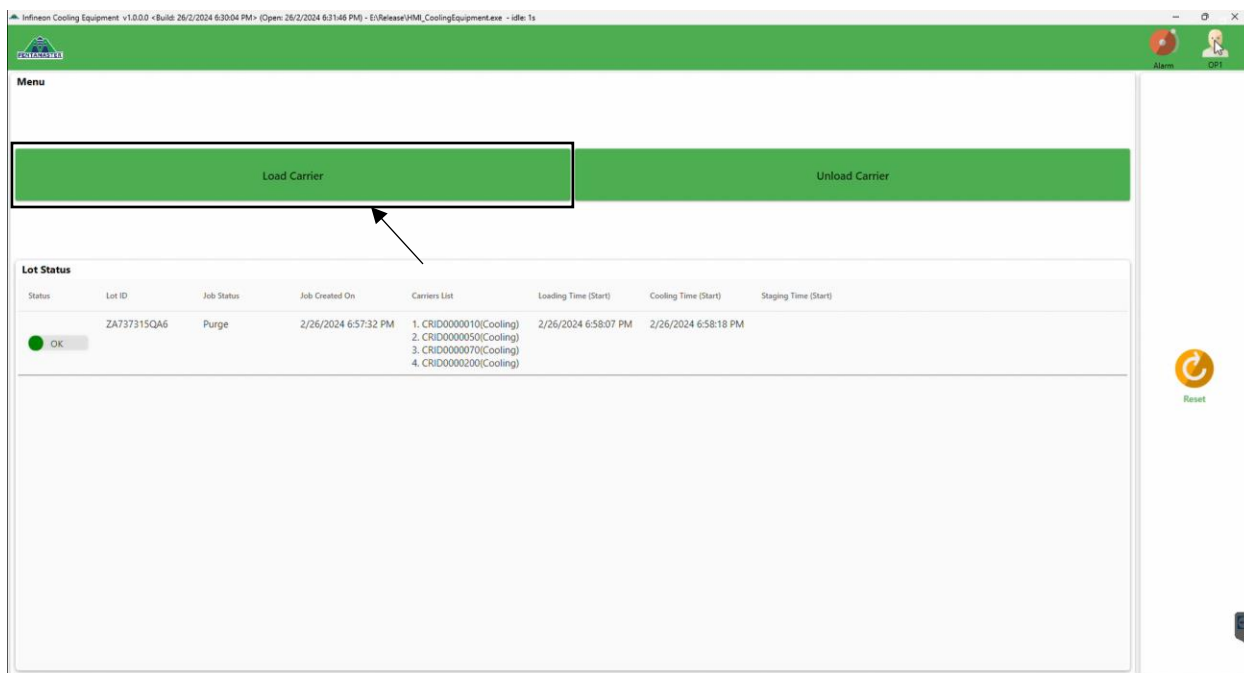
3.2.1 Loading Process

1. Login into the system.

Before initiating any operation, the operator must log into the system using their designated credentials. Refer to section 3.1.1 for instructions on accessing the login interface.

2. Click on “Load Carrier” button.

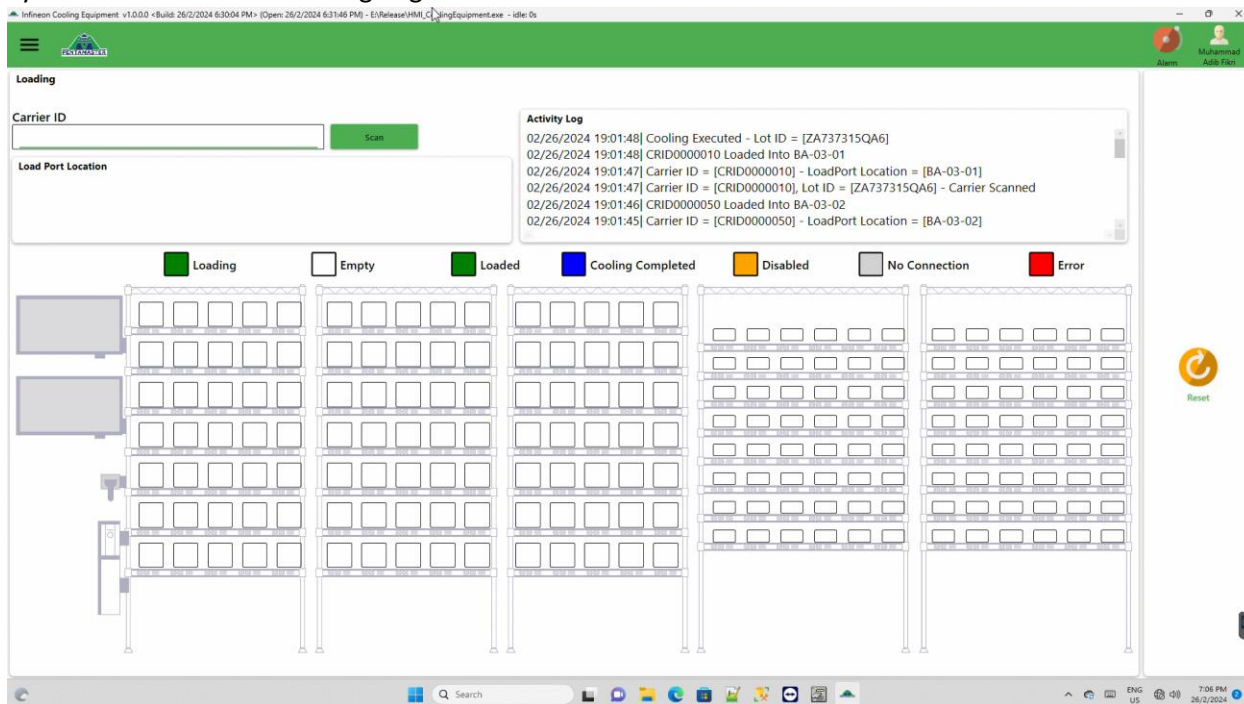
Clicking on this button signifies the initiation of the carrier loading process.





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System will redirect to Loading Page.



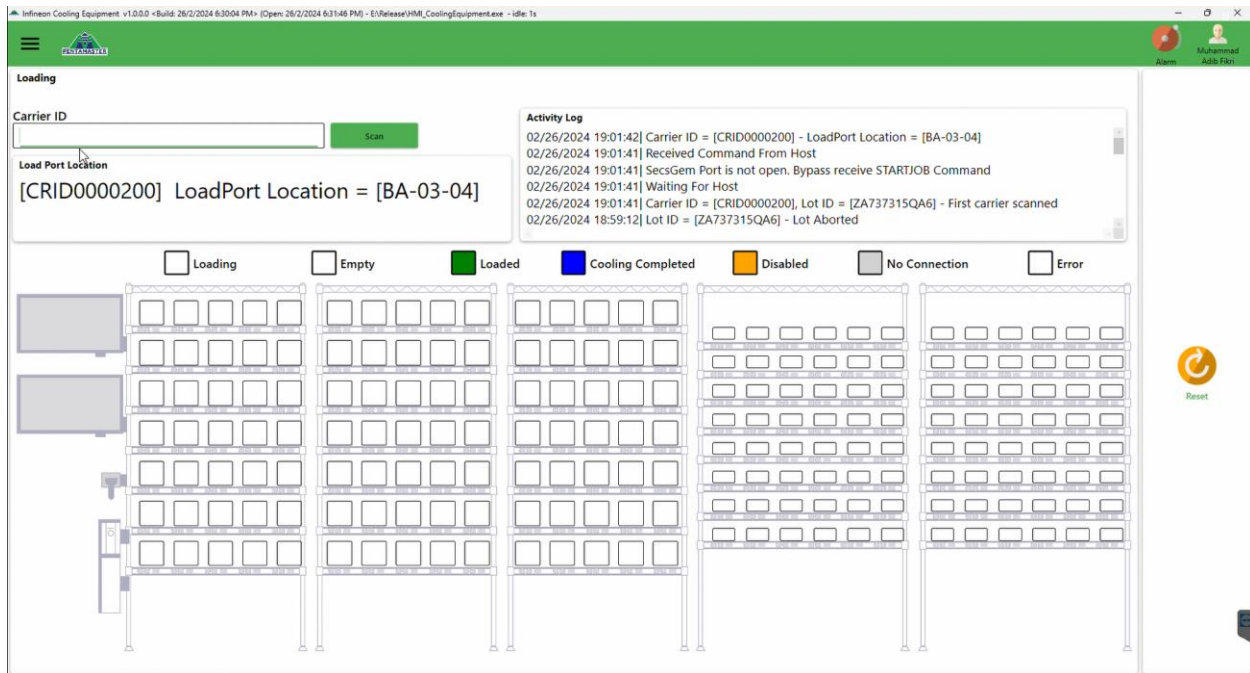
3. Scan Carrier ID

Utilizing the integrated barcode scanner, scan QR code affixed to the carrier.





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4. Load Port Light Indicator

Upon successful scanning, the load port light indicator will start blinking green, indicating readiness to receive the carrier.



5. Load Carrier into Load Port

Physically place the carrier into the designated load port of the system, ensuring proper positioning for further processing.



6. Confirmation of Loading

Once the carrier is securely loaded, the load port light indicator will change to a solid green status, confirming successful detection and loading by the system.



7. Repeat Steps 3 to 6

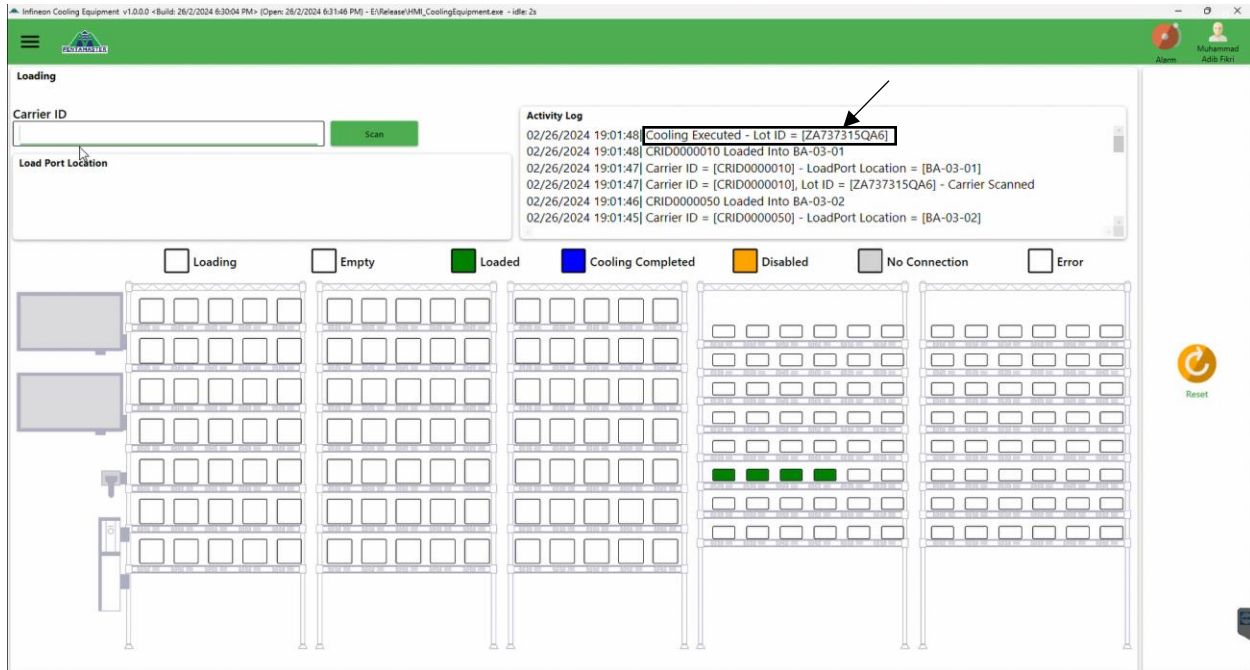
Repeat the scanning and loading process for each subsequent carrier until all carriers have been loaded into the system.



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8. Activity Log Confirmation

Upon completion of carrier loading, the system's activity log will display a "Cooling Executed" message, confirming successful execution of the operation.



9. Logout from the system.

After completing all necessary operations, ensure to log out from the system to maintain security and prevent unauthorized access. Refer to section 3.1.2 for logout instructions.

3.2.2 Unloading Process

1. Login into the system.

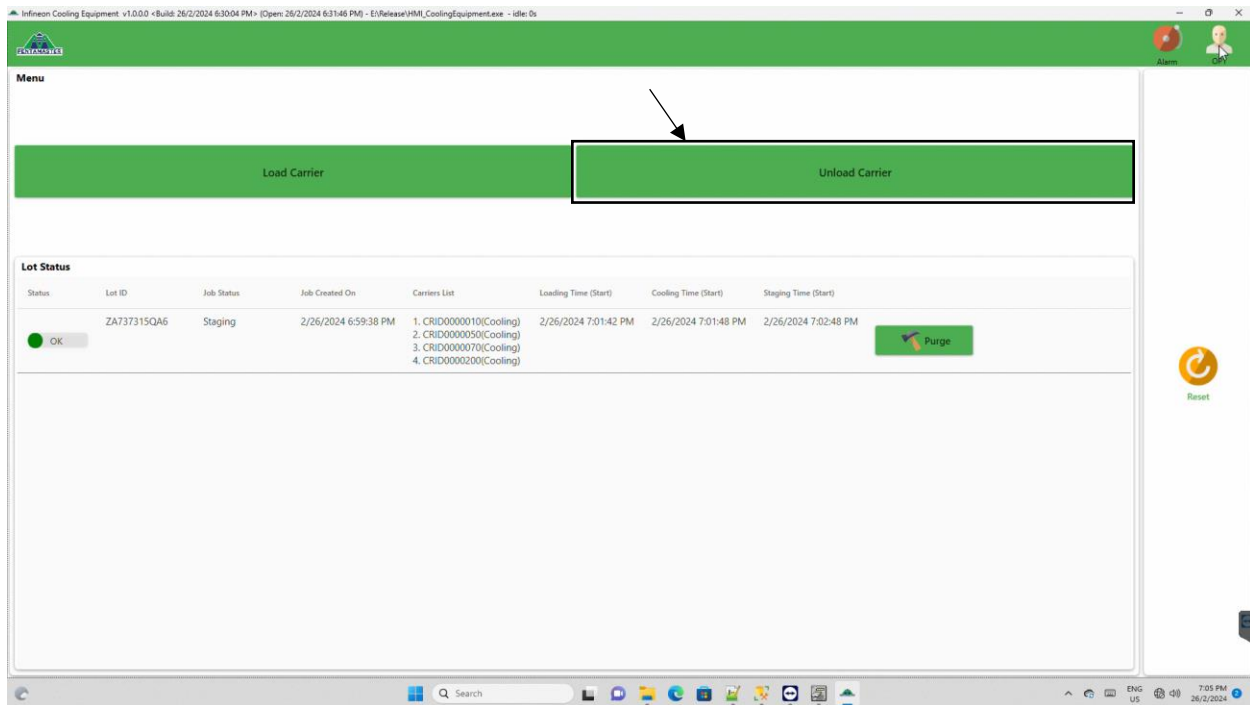
Before initiating any operation, the operator must log into the system using their designated credentials. Refer to section 3.1.1 for instructions on accessing the login interface.

2. Operator click on "Unload Carrier" Button.

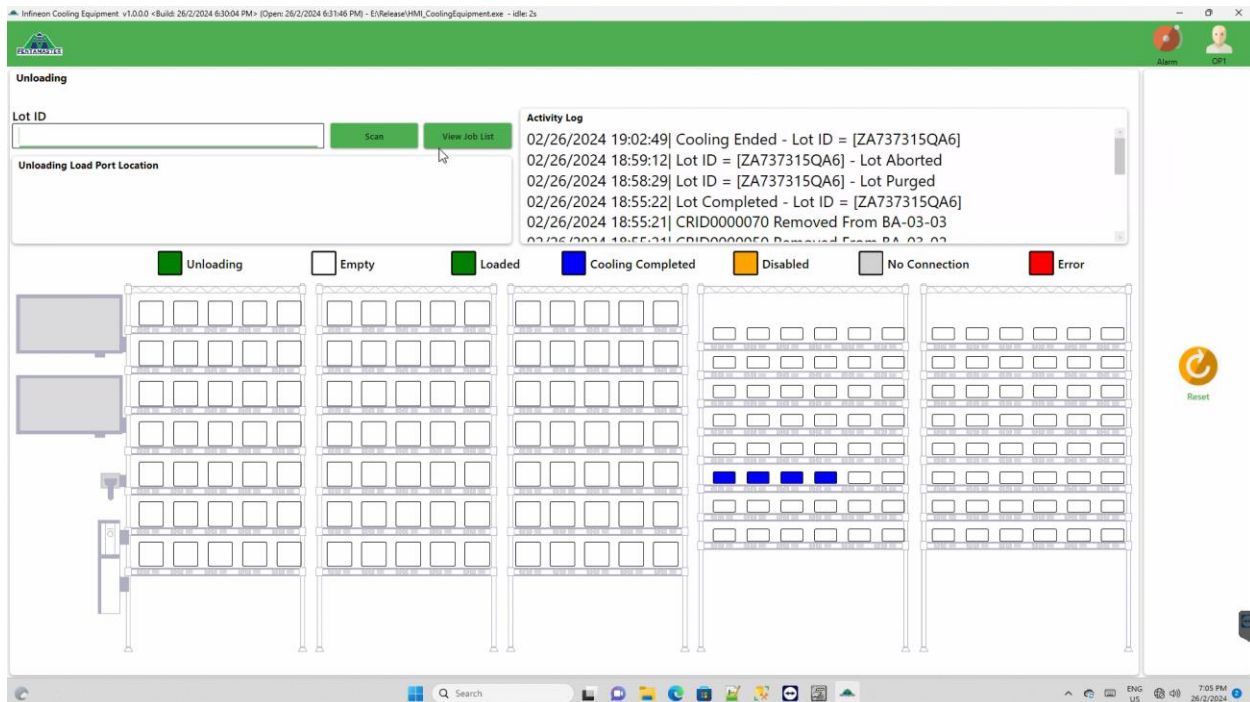
Clicking on this button signifies the initiation of the carrier unloading process.



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System will redirect to Unloading Page.



3. Scan Lot ID

Utilizing the integrated barcode scanner, scan the barcode or QR code associated with the lot.

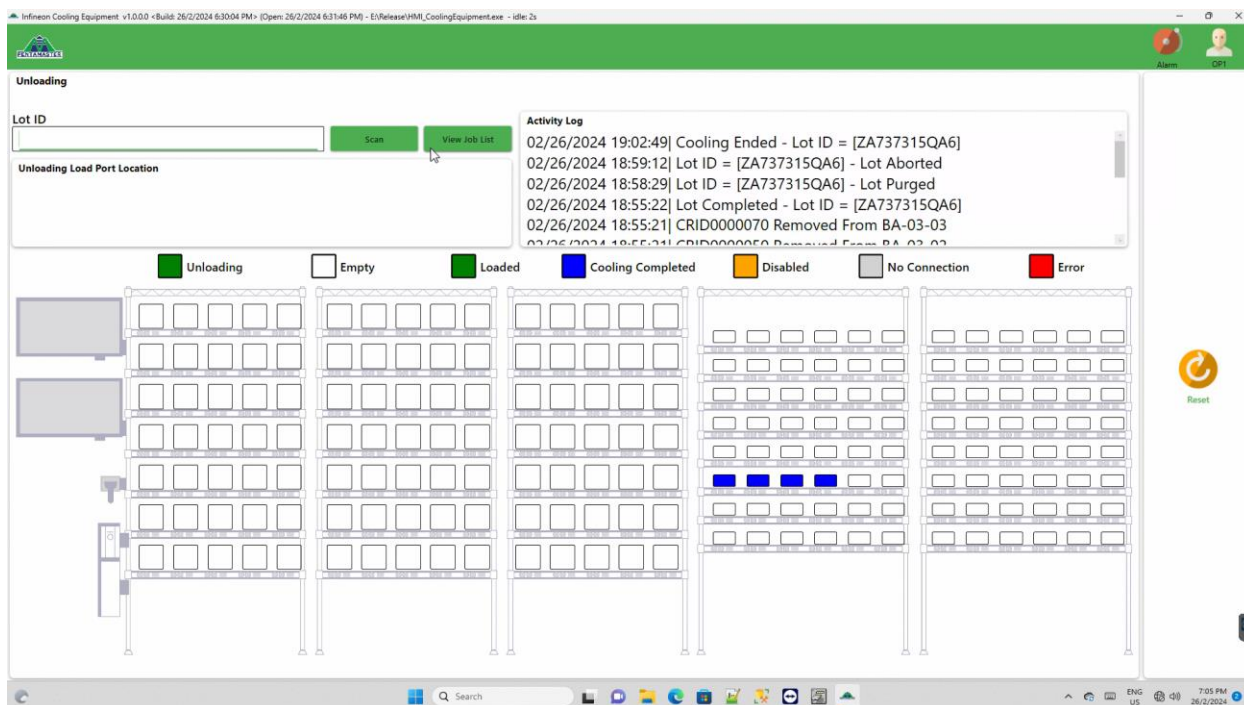


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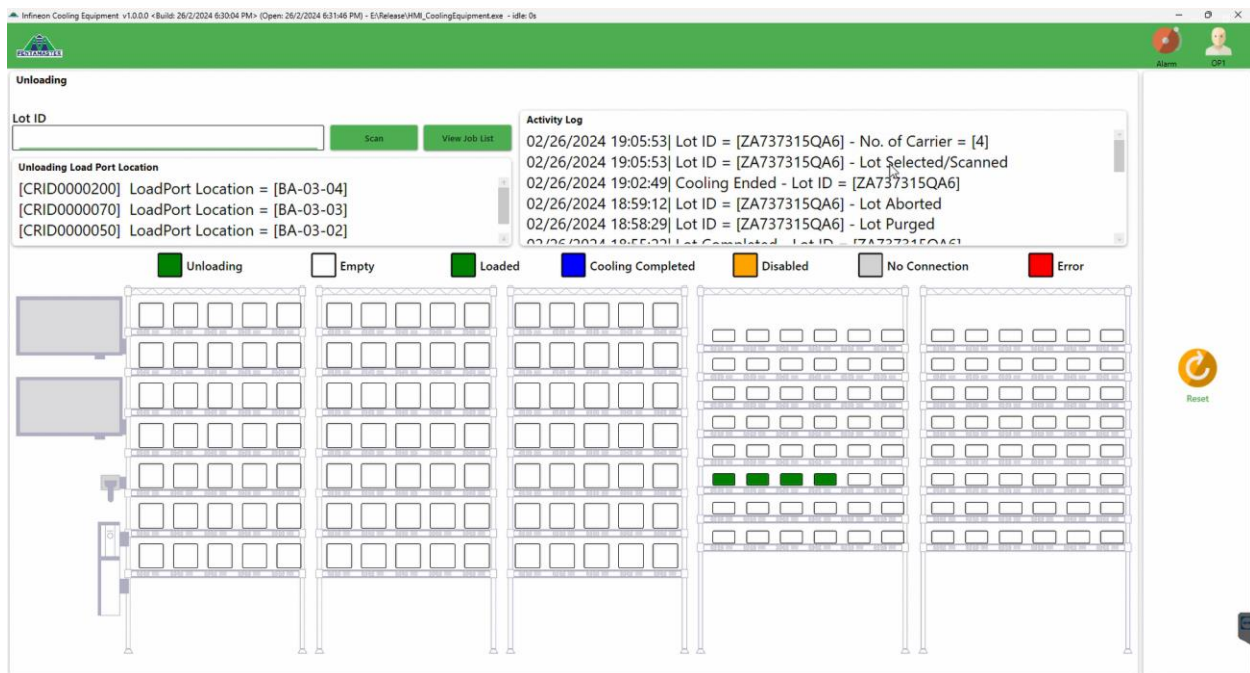
4. Load Port Light Indicator

Upon successful lot ID scanning, the respective load port light indicator will change color from solid blue to blinking green, indicating readiness for carrier unloading.





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5. Unload Carrier from Load Port

Physically remove the carrier from the designated load port of the system. Ensure safe handling and proper extraction to avoid any potential damage.



6. Load Port Light Indicator Status



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Once the carrier is successfully unloaded, the load port light indicator will turn off, signaling that the port is now vacant.



7. Repeat Steps 5

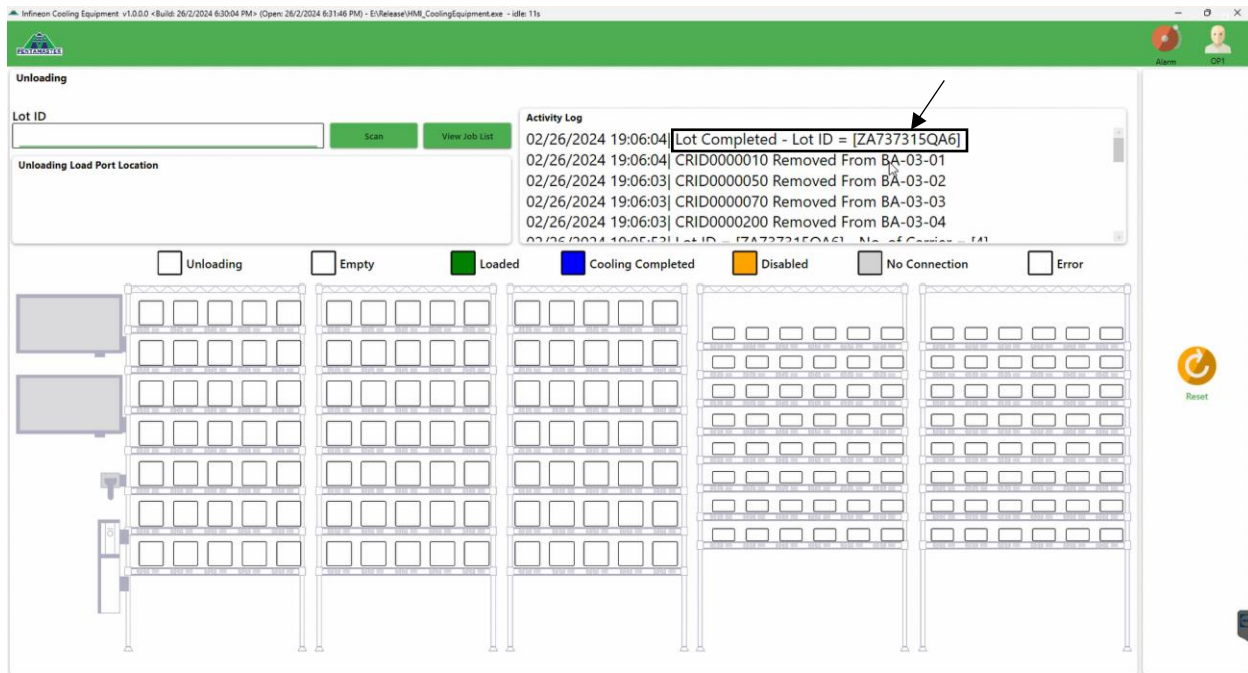
Repeat the unloading process for each subsequent carrier until all carriers within the lot have been successfully unloaded from the system.

8. Activity Log Confirmation

Upon completion of carrier unloading, the system's activity log will display a "Lot Completed" message, confirming the successful unloading of all carriers within the lot.



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9. Operator Logout

After completing all necessary operations, ensure to log out from the system to maintain security and prevent unauthorized access. Refer to section 3.1.2 for logout instructions.

3.2.3 Purge and Abort Job

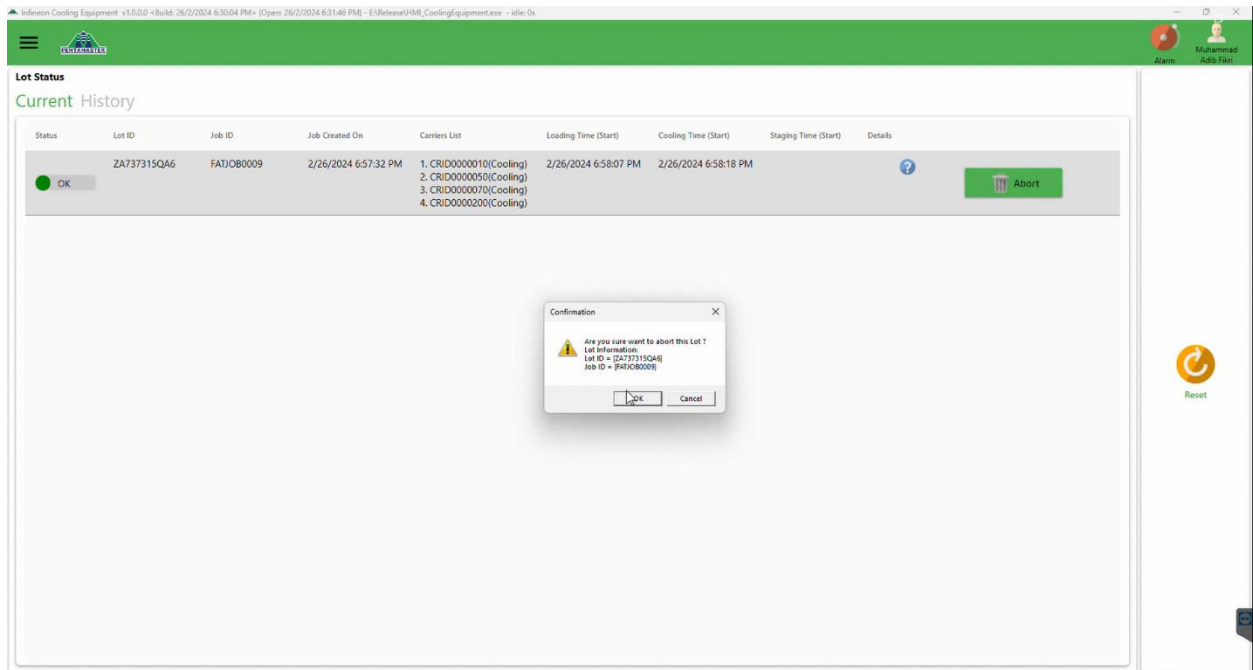
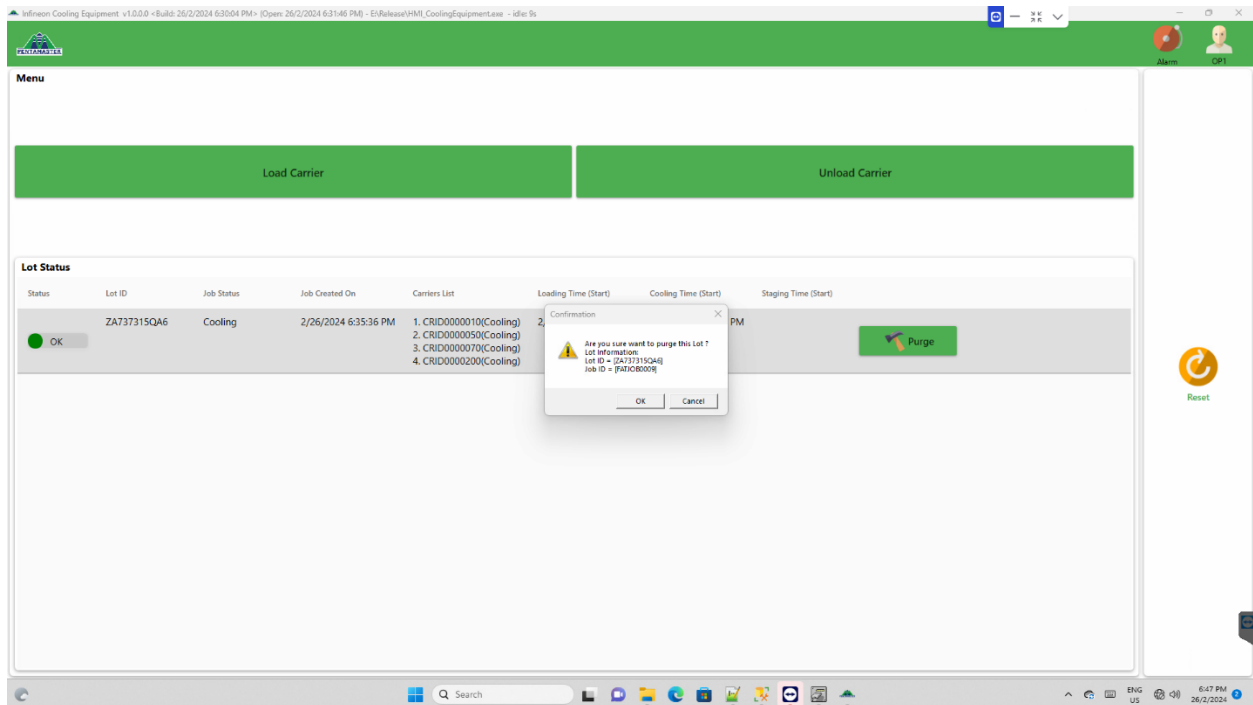
To utilize the purge and abort job feature:

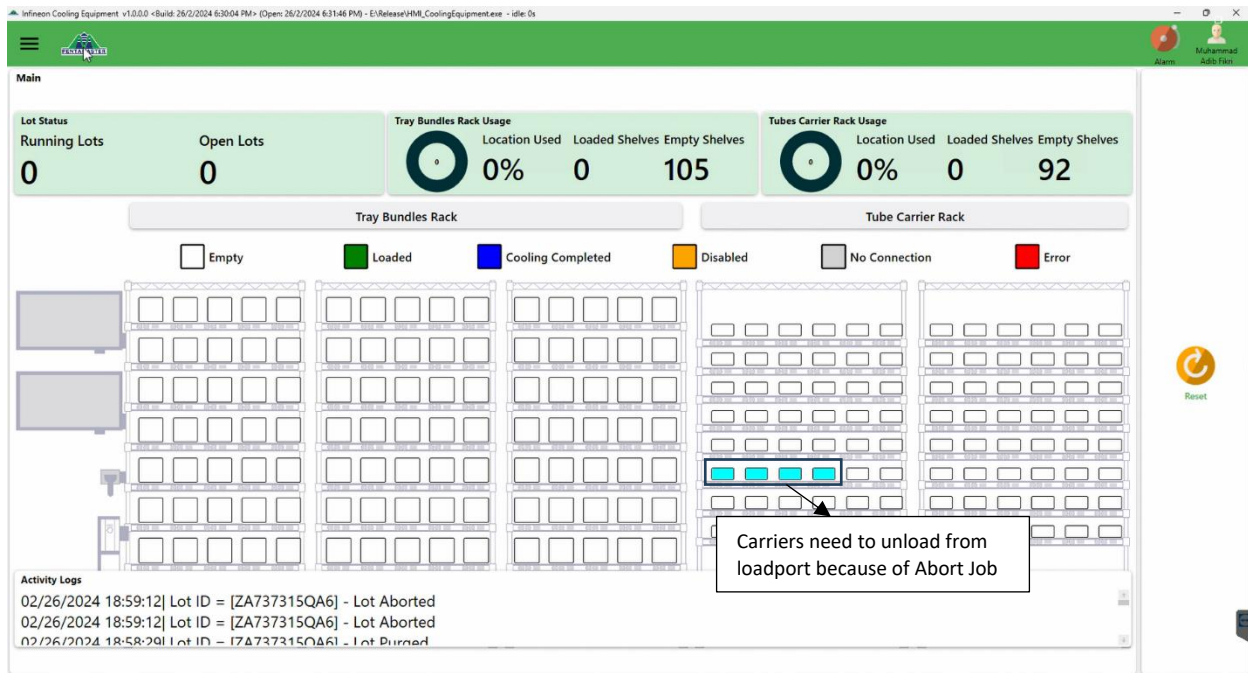
1. Click on the "Purge" button associated with the relevant job to notify the HMI system to ignore the job. Confirm the action by clicking "OK" to proceed.
2. If an operator initiates the purge, they must then request an engineer to log in and abort the job. Upon triggering the abort job process by the engineer, any related loadports associated with carriers from that job will blink light blue, indicating their readiness for unloading.

This functionality allows for efficient management of errors encountered during operation, ensuring minimal disruption to equipment functionality while awaiting resolution by engineering staff.



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3.2.4 Handling Alarms and Warnings

In this section, we will discuss how to handle alarms and warnings within the HMI system.

If alarms or warnings occur:

Users can attempt the suggested action to resolve the issue and clear the alarm or warning.

Alternatively, users can click on the reset button to clear the alarm or warning.

To identify the source of the alarm or warning:

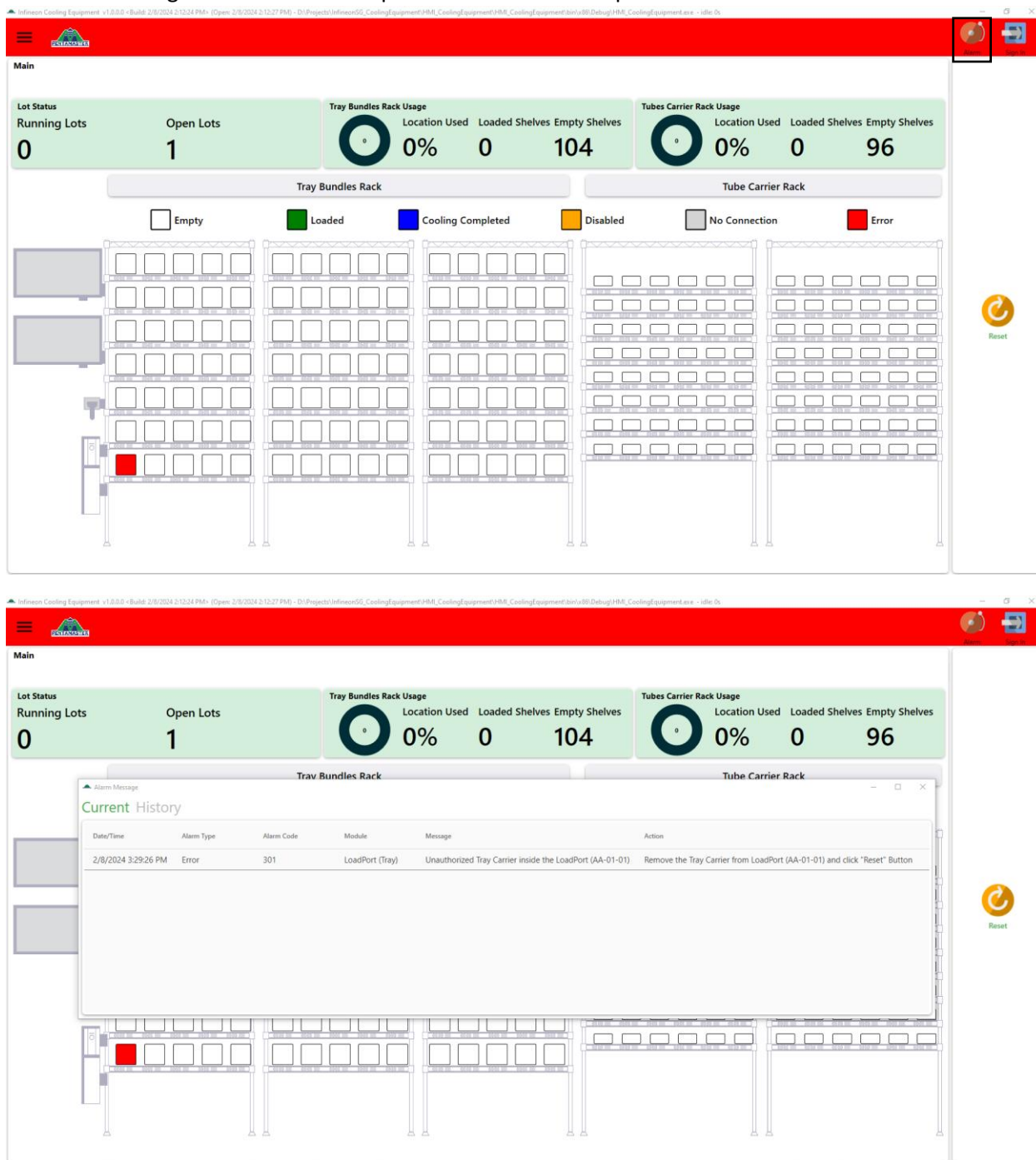
Click on the "Alarm" button located at the top right side of the interface.

The system will open a window displaying all current alarms and warnings.



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Refer to the image below for an example of an alarm on load port AA-01-01.



3.2.5 Enabling/Disabling Load Ports

To enable or disable load ports within the HMI system, follow these steps:

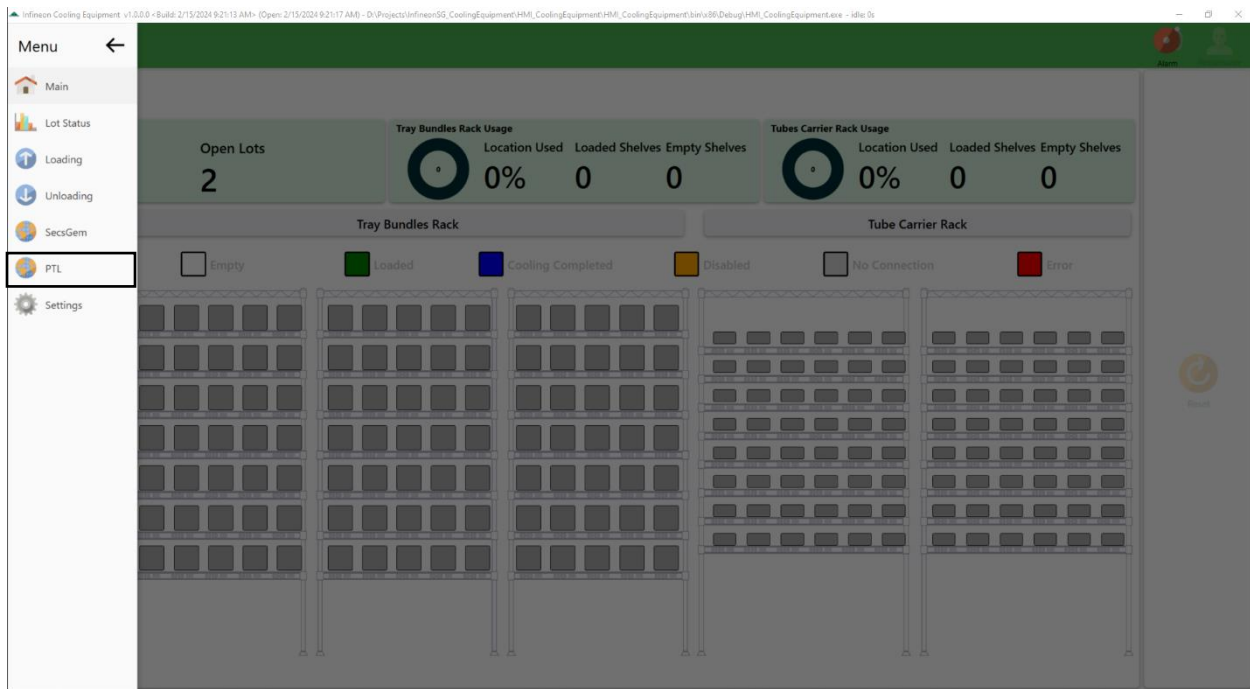
1. Ensure that you are logged in with a user level other than "Operator". If you are logged in as an Operator, you will not have access to this functionality.



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2. Once logged in, navigate to the menu located on the top left side of the interface and select “PTL”.
3. The system will then redirect you to the PTL Page, where you can manage load port settings.
4. On the PTL Page, locate the checkbox labeled “Enable”. Check this checkbox to enable the load port or uncheck it to disable the load port.
5. After making your selection, ensure to save any changes made.

By following these steps, you can effectively enable or disable load ports as needed within the HMI system.





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